

Avantika Agarwal

agarwalavantikaint@gmail.com | [linkedin.com/in/agarwalavantika](https://www.linkedin.com/in/agarwalavantika) | github.com/agarwalavantika | agarwalavantika.github.io

Summary

Machine Learning researcher developing robust speaker verification and speech representation systems for pathological and multilingual speech. Currently building interpretable dysarthria assessment methods using self-supervised speech models (wav2vec2, WavLM, XLS-R, IndicWav2Vec), DTW-aligned phonetic posteriorgrams, and deep speaker embeddings (ECAPA-TDNN, CAM++), evaluated on public dysarthric speech corpora (UA-Speech) with cross-lingual and Hindi-specific analysis. Interested in building reliable, clinically interpretable AI systems for healthcare and accessibility.

Publications

- Interpretable Features for Dysarthria Assessment via DTW-Aligned Phonetic Posteriorgrams — Submitted. Five interpretable biomarkers from DTW-aligned wav2vec2 posteriorgrams achieve AUC 0.952 on UA-Speech (28 speakers, 21,380 utterances), statistically matching a 1024-dim wav2vec2 embedding baseline (AUC 0.970) and outperforming an MFCC-SVM baseline (AUC 0.917).
- Encoding Articulatory Features in SSL Models for Hindi Dysarthric Speech — Submitted, ISCSLP 2026. Linear probing of WavLM, XLS-R, and IndicWav2Vec on isolated Hindi vowels/consonants (13 healthy, 14 dysarthric speakers) shows voicing survives dysarthria best while place and manner retain the most usable signal after chance-correction; identifies collapse of Hindi-specific contrasts (retroflex/dental, aspiration).
- Prosodic Features for Classification of Normal vs. Dysarthric Speech — Accepted, SPECOM 2026.
- Neural Similarity Learning for Dysarthric Cross-Lingual and Multilingual Speaker Verification — Submitted, SLT Workshop 2026.
- Reproducibility Study — Speech Model Evaluation — Submitted, TMLR (Reproducibility Certification).

Research Interests

Speaker Verification & Speaker Embeddings • Pathological Speech Processing (Dysarthric Speech) • Self-Supervised Speech Representation Learning (wav2vec2, WavLM, XLS-R) • Cross-Lingual & Multilingual Representation Learning • Interpretable / Explainable Speech AI • ASR/TTS Systems • AI for Healthcare and Accessibility

Education

M.Tech in ICT (Machine Learning) — *DA-IICT, Gandhinagar, India* Jul 2024 – Jul 2026
CPI: 8.03

B.Tech in Computer Science (First Division with Honours) — *Govind Ballabh Pant Engineering College, India* Aug 2020 – Jun 2024
Percentage: 82%

Research Experience

Research Intern — *Speech Research Lab, DAU* Feb 2026 – Present

- Developed a DTW-aligned phonetic-posteriorgram framework over wav2vec2-CTC representations, deriving five clinically interpretable biomarkers for dysarthria assessment on the UA-Speech corpus (28 speakers, 21,380 utterances).
- Achieved AUC 0.952 with the interpretable biomarker set under 5-fold GroupKFold CV — statistically competitive with a 1024-dim wav2vec2 embedding baseline (AUC 0.970, SE=0.024) and outperforming an MFCC-SVM baseline (AUC 0.917), validated via ablation, likelihood-ratio testing, and speaker-level DeLong testing.
- Probed WavLM, XLS-R, and IndicWav2Vec SSL representations on isolated Hindi vowels/consonants (13 healthy, 14 dysarthric speakers) via linear probing with leave-one-speaker-out CV, decomposing consonants into place, manner, voicing, and aspiration.
- Identified that Hindi-specific phonological contrasts (retroflex/dental place, four-way aspiration) measurably collapse under dysarthria, with Hindi-aware pretraining (IndicWav2Vec) improving chance-corrected residual signal.
- Implementing deep speaker embedding architectures including ECAPA-TDNN and CAM++, with a neural back-end model for learned similarity estimation beyond cosine scoring.
- Performed monolingual and cross-lingual speaker verification experiments to analyze domain generalization and robustness to distribution shift from healthy speech.

- Developed reproducible PyTorch pipelines for feature extraction, training, and evaluation.

AI Researcher, NLTMT Bhasini Project — *MeitY, Government of India*

Jan 2026 – Jun 2026

- Worked as project staff under the National Language Translation Mission (Bhasini), contributing to ASR/TTS systems as part of a national-scale language technology initiative.
- Supported development and evaluation of speech systems across Indian languages.

Machine Learning Intern — *Radiant Logic (Remote)*

Dec 2025 – Feb 2026

- Researched transformer-based Text-to-SQL systems for enterprise knowledge retrieval.
- Designed synthetic data generation pipelines improving schema alignment and long-context reasoning.
- Explored uncertainty-aware inference and evaluation strategies for production-scale ML systems.

Prompt Engineer (Contract) — *Turing (Remote)*

Feb 2025 – Oct 2025

- Developed and evaluated LLM workflows including LoRA-based fine-tuning and prompt optimization.
- Assisted in dataset preparation, benchmarking, and inference evaluation for large language models.

Research Mentee — *Amazon ML Summer School*

Sept 2023 – Nov 2023

- Selected for Amazon ML Summer School focusing on modern machine learning research methodologies.
- Studied transformer architectures, representation learning, and large-scale model training.
- Explored reinforcement learning and retrieval-augmented generation pipelines.
- Gained formal training in experimental design, model evaluation, and research problem formulation.

Research Collaborator — *ESOC Research Program (Time Series Forecasting)*

Mar 2024 – Jul 2024

- Conducted comparative research on statistical and deep learning approaches for time-series forecasting.
- Benchmarked ARIMA, Prophet, and LSTM models using rolling-window evaluation protocols.
- Investigated temporal drift and robustness of forecasting models under non-stationary conditions.

Selected Research Projects

Neural Backend Speaker Verification for Dysarthric Speech — *Current Work*

- Proposed a neural backend model for learned similarity estimation in speaker verification, using ECAPA-TDNN embeddings in place of cosine scoring.
- Designed feature representations combining speaker embedding differences and interaction features.
- Demonstrated consistent performance improvements over cosine similarity baselines under cross-lingual verification conditions.

Interactive Image Segmentation with Click-Guided Deep Networks —

- Developed an interactive segmentation framework using a modified U-Net (MFP-ResNetUNet) with a pre-trained ResNet-34 encoder.
- Extended network input to a 5-channel representation combining RGB with simulated foreground/background click maps.
- Achieved Dice score of 0.7994 and IoU score of 0.6659 on the BSDS500 evaluation set.

Technical Skills

- **Speech & Audio Models:** wav2vec2 (incl. wav2vec2-CTC), WavLM, XLS-R, IndicWav2Vec, ECAPA-TDNN, CAM++, Speaker Embeddings, Phonetic Posteriorgrams, MFCC, Log-Mel Spectrograms, ASR, TTS, Pathological/Dysarthric Speech Processing
- **Speech Datasets:** UA-Speech, Hindi Dysarthric Speech Corpus (isolated vowels/consonants)
- **Machine Learning:** Speaker Verification, Neural Similarity Learning, DTW Alignment, Linear Probing, Transfer Learning, Cross-Lingual & Multilingual Modeling, Statistical Evaluation (AUC, DeLong's test, ablation)
- **LLMs:** Transformers, LoRA, PEFT, Retrieval-Augmented Generation (RAG), Prompt Optimization
- **Frameworks & Tools:** PyTorch, HuggingFace, Scikit-learn, Librosa, NumPy, Pandas
- **Programming:** Python, C++, SQL, Shell
- **Computer Vision:** U-Net, ResNet, Interactive Segmentation, GAN-based Architectures, Image Representation Learning

Achievements

- GATE CSE 2024: 96.46 percentile (Top 3%)
- Qualified UGC NET (PhD Eligibility, Computer Science & Applications)
- Harvard WEAmplify Scholar
- Microsoft Learn Student Ambassador (Beta)